

# Luwei Wang

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## Profile

ML researcher and PhD candidate in Biomedical AI at the University of Edinburgh, working on **scalable Bayesian inference**, **causal machine learning**, and **efficient LLM reasoning**. First-author papers at **ICML 2026** and **UAI 2025** on Bayesian clustering for high-dimensional and longitudinal data, plus a co-authored **ICML 2026** paper on LLM test-time inference.

## Education

**University of Edinburgh** 2023 to 2027 (expected)  
*PhD in Biomedical AI, UKRI CDT* Edinburgh, UK

- Advisor: Dr Sohan Seth at the Data Science Unit, School of Informatics.
- Focus: Bayesian modelling, semi-supervised and causal clustering, LLM test-time scaling.

**University of Edinburgh** 2021 to 2022  
*MSc in Statistics with Data Science, Distinction* Edinburgh, UK

- Dissertation: Clustering Individual Trajectories of Multiple Long-term Conditions. Clustered longitudinal GP records from around 150k UK Biobank participants to identify multimorbidity phenotypes and predict adverse outcomes, using a K-means and RNN framework with attention-based temporal feature extraction.

**Hong Kong Baptist University** 2017 to 2021  
*BSc (Hons) Mathematics and Statistics, First Class. Minor in Computer Science* Hong Kong

## Selected Publications

- **Wang, L.**, Panas, D., Wang, K., Guthrie, B., Seth, S. *Scalable Bayesian Semi-supervised Clustering with Feature Selection and Adaptive Constraint Weighting*. **ICML 2026**.
- Wang, K., Yu, Z., **Wang, L.**, Huang, Y. *Cache Coherent Resampling for Efficient Test Time Scaling in LLM Reasoning via Adaptive Sequential Monte Carlo*. **ICML 2026**.
- **Wang, L.**, Lone, N., Seth, S. *Bayesian Supervised Causal Clustering*. arXiv:2603.05288, 2026.
- **Wang, L.**, Richards, K., Seth, S. *Nonparametric Bayesian Multi-Facet Clustering for Longitudinal Data*. **UAI 2025**, PMLR 286, 4411 to 4442.

## Research Experience

**Research Assistant** Oct 2022 to Jan 2023  
*Data Science Unit, School of Informatics, University of Edinburgh* Edinburgh, UK

- Implemented and benchmarked supervised clustering methods (Supervised K-means, Mahalanobis metric learning, and VAE-based models) on cytokine biomarker data from critically ill COVID-19 patients.
- Introduced a *FavorCP* outcome that improved odds-ratio testing across discovered subgroups under convalescent plasma treatment, motivating the fully Bayesian causal formulation later published as BSCC.
- Delivered a Turing-funded technical report on subphenotype discovery for personalised treatment.

## Selected Projects

**ASMC: Adaptive Sequential Monte Carlo for LLM Test-Time Scaling** *ICML 2026*

- **Training-free** parallel SMC for sampling power-shaped trajectory distributions in LLM reasoning, replacing chain-based MCMC. Runs on a fixed base LLM with no fine-tuning or RL, using residual resampling, batched decoding, and adaptive compute allocation.
- Cache-coherent resampling reorders KV caches under the ancestor mapping, avoiding  $O(NL^2)$  prefix-replay attention cost and keeping resampling overhead small even beyond 3k tokens.
- On MATH500 at  $128\times$  budget, ASMC attains **80.6%** exact-match accuracy with p95 latency **73.7s**, versus **1318s** for chain-based MCMC. Improves the accuracy/p95 Pareto trade-off over both MCMC and best-of-N baselines.

**BASIL: Scalable Bayesian Semi-supervised Clustering** *ICML 2026*

- Built a Bayesian semi-supervised clustering model on an HMRF with stochastic variational inference, jointly learning cluster assignments, cluster-specific feature relevance, and adaptive latent constraint weights.
- Trained on MNIST (**60k** binarised samples) with supervision in  **$\sim 1.5$  hours**, while metric-learning baselines (MML, PCK, MPCK) fail to converge in over 2 days, a  **$>96\%$**  compute-time reduction. Scaled further to ChestMNIST and a UK CPRD cohort of **501k** patients with **61** binary condition indicators.
- Adaptive constraint weighting maintained accuracy under up to **30% noisy** supervision; recovered clinically interpretable multimorbidity subgroups.

**BSCC: Bayesian Supervised Causal Clustering** *arXiv 2026*

- Designed a Bayesian framework that uses individual treatment effects as the outcome to guide clustering, recovering subgroups homogeneous in both covariate profile and treatment response.
- Joint generative model over covariates, treatment, and outcome under unconfoundedness, yielding interpretable subgroups not surfaced by unsupervised clustering or counterfactual regression baselines.
- Validated on simulated benchmarks and the third International Stroke Trial cohort. [\[arXiv\]](#)

#### **NPBayes-MFC: Nonparametric Bayesian Multi-Facet Clustering for Longitudinal Data** *UAI 2025*

- Developed a nonparametric Bayesian framework that fits a separate clustering for each facet of multi-faceted longitudinal data, with nonparametric priors selecting the number of clusters automatically.
- Derived a Variational Bayesian inference scheme over nonlinear growth and vector autoregressive likelihoods, enabling scalable training where MCMC-based baselines are intractable. First implementation of multi-facet clustering for nonlinear growth models.
- Applied to income trajectories from the English Longitudinal Study of Ageing (ELSA), surfacing interpretable subgroups that single-facet baselines collapse together. [\[Paper\]](#)

### **Technical Skills**

**ML and probabilistic programming.** PyTorch, HuggingFace Transformers, NumPyro, Pyro, Stan, scikit-learn.

**Methods.** Variational inference, MCMC and sampling methods, nonparametric Bayes, causal inference and HTE, semi-supervised and metric learning, deep generative models, LLM reasoning and test-time scaling.

**Engineering.** Multi-GPU training, Git, Linux and SLURM clusters, Docker, LaTeX.

**Programming languages.** Python, R, SQL, MATLAB, Java.

**Spoken languages.** Mandarin (native), English (fluent).